

According to a press report by the company, the flex-cells produced by this technology can be cut to order, from a module five metres long to the size of a credit card. The uniqueness of the product is that it remains flexible—not just the solar photovoltaic module but even after encapsulation.

Another technology innovator worth a mention in this space is Oder-sun, whose solar cells can be created in any size or dimension, as was noted in the case of trapezoidal modules used to form concentric rings that powered the visitor centre buildings of the Olympic Park in Beijing in 2008! This technology is now coming to the aid of architects developing customised roofing solutions, especially in Germany.

It is believed that nanotechnology is going to cause the next disruptive change in solar power panels. Nanotechnology has the potential to create circuits out of individual silicon molecules, which may cost half as much as traditional PV cells.

In India, Applied Materials helped set up a Nanotechnology Research Center at the Indian Institute of Technology, Bombay to promote better understanding of nanofabrication methods, materials and processes for applications in semiconductors and solar.

A company called Nanosolar in San Jose, California, has secured more than \$100 million from investors to build a factory for nanotechnology-based thin-film solar panels.

Then, there are others trying to marry the benefits of crystalline and thin-film technologies. Sanyo's heterojunction with intrinsic thin-layer (HIT) solar cell is an example. This hybrid combines a crystalline silicon substrate and amorphous silicon thin-film. To date, it offers the world's best power generation level per unit of installation area, based on superior high energy conversion efficiency and temperature resistance. Recently, Sanyo demonstrated a HIT cell with a thickness of just 98 micrometres—half that of the earlier solar cell. This comes with 22.8 per cent energy conversion efficiency of the cell.

Green certificates, now in India

In September this year, the Indian government announced that distribution companies have to replace 6 per cent they distribute with alternative energy like solar, wind or hydroelectric power. They have also created a new financial instrument called the reduction in emission certificate or REC, which will be sold by Indian companies producing renewable energy to distribution companies. One REC will represent 1 megawatt-hour (MWh) of power. These RECs can be bought by distribution companies in regions that have nil or weak potential to produce alternative power. RECs help such distribution companies to indirectly meet their renewable energy obligations!

With India's 160,000MW installed capacity, 9600 MW has to be bought from renewable energy companies physically or as REC. Considering the government-established price band of Rs 1.5-3.67/MWh, the market is worth a minimum Rs 144 billion. A huge business opportunity indeed, but it needs eager buyers!

Another interesting breakthrough is Shrink Nanotechnologies' shrinkable plastic film. In a press report, Sayantani Ghosh, consultant to Shrink Nanotechnologies, explained that this technology is a completely unique process: "In a solar cell you take sunlight, and convert it into electricity. What we are doing is taking sunlight and converting it into light of a different colour. This different coloured light then falls onto the existing silicon PV. The colour of the light is set to the PV's preferred colour. It is like straining the sunlight into colours that will enhance the efficiency of the silicon." This would enable homes to install a thin solar concentrator between the glass sheets in a window, and the collected energy can be converted into electricity thereafter.

On the device front...

The many innovations discussed in the previous section are enabling solar-powered devices ranging from wireless sensors to mobile phones.

In mid-2010 Vodafone revealed a solar-powered mobile phone for the Indian market, priced at Rs 1500. Samsung debuted its solar handset in India last year. There are generic solar-powered backup batteries that can be used with multiple brands. Some companies have also demonstrated solar power backup systems for laptops. These range in form from solar panel stickers that can be stuck on the top of a laptop to bulky charging panels. They are mostly offered by small start-ups like SolarRolls and Sunshine Solar,

and are yet to become common or even available in India.

Although Nokia has not focused much on solar, it has unveiled other interesting prototypes such as bicycle-pedal-powered chargers and radio-wave-powered cellphones!

For Google, the Sun is the limit—or perhaps not

The Internet-Czar Google has been investing in clean and green energy research for a few years now. In March, it announced that it has developed a new mirror or heliostat which could halve the cost of building a concentrated solar power (CSP) plant.

In general, solar thermal projects use mirrors to focus the sun's rays to heat a substance, which, in turn, produces steam to run a turbine. Google has been researching with new materials and substrates to produce low-cost, low-maintenance, mass-producible heliostats that would focus the sun's beams in a more efficient way.

Google is likely to join hands with CSP companies like eSolar and BrightSource to bring this technology to the market within two to three years.

Once available commercially, this technology is likely to reduce the cost of setting up 'solar power towers.' A solar power tower is a type of solar furnace that uses a tower to receive the sunlight focused onto it by heliostats. The focused light is used to heat up water or liquid sodium, to produce enough steam to run turbines. There are a few functional solar power towers such as